

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL1- INDUSTRY PROBLEM SOLVING TASK
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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Department:	AIML	Date:	21.08.26

ANNEXURE A
INDUSTRY PROBLEM SOLVING TASK

1. Smartphone Performance Optimization

A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

Smartphone Performance Optimization

Problem: A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

Answer

The lag occurs because frequently used application data must move between the processor and slower DRAM. A suitable memory hierarchy can reduce this delay.

Memory	Role	Latency	Capacity
L1 Cache	Stores most frequently used instructions/data	Very low	Small
L2 Cache	Stores recently used data not present in L1	Low	Medium
L3 Cache	Shared cache for CPU cores	Higher than L1/L2	Large
DRAM	Stores active applications and OS data	High	Very large

Proposed Redesign

CPU → L1 → L2 → L3 → DRAM → Storage

- Increase L1 cache moderately for frequently accessed data.
- Use larger L2 cache to reduce DRAM accesses.
- Increase shared L3 cache for efficient communication between CPU cores.
- Use high-speed LPDDR DRAM with multiple channels.
- Apply cache prefetching to predict application data.
- Use memory compression to increase effective DRAM capacity.

Result

The redesigned hierarchy reduces DRAM accesses, lowers memory latency and improves application switching and overall throughput.

2. Cloud Server Memory Bottleneck

A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.

Autonomous Vehicle Data Processing

Problem: An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

Answer

Autonomous vehicles generate large amounts of camera, radar and LiDAR data. The memory system must provide high bandwidth with very low latency.

Memory	Advantage	Limitation
SRAM	Extremely fast, low latency	Expensive and low capacity
DRAM	High capacity and good bandwidth	Higher latency, volatile
MRAM	Non-volatile, fast and low power	More expensive than DRAM

Proposed Hierarchy

CPU/AI Accelerator → SRAM/L1 → L2/L3 Cache → DRAM → MRAM

- Use SRAM for real-time sensor processing.
- Use L2/L3 caches for frequently accessed intermediate data.
- Use high-bandwidth DRAM for large sensor datasets.
- Use MRAM for persistent parameters, configuration and critical data.
- Use parallel memory channels to increase bandwidth.
- Use DMA to transfer sensor data without heavily loading the CPU.

Result

This hierarchy provides low latency for critical data, high bandwidth for sensor processing and non-volatile storage using MRAM.

3. Autonomous Vehicle Data Processing

An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

Autonomous Vehicle Data Processing

Problem: An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

Answer

Autonomous vehicles generate large amounts of camera, radar and LiDAR data. The memory system must provide high bandwidth with very low latency.

Memory	Advantage	Limitation
SRAM	Extremely fast, low latency	Expensive and low capacity
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Proposed Hierarchy

CPU/AI Accelerator → SRAM/L1 → L2/L3 Cache → DRAM → MRAM

- Use SRAM for real-time sensor processing.
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Result

This hierarchy provides low latency for critical data, high bandwidth for sensor processing and non-volatile storage using MRAM.

4. AI Inference Edge Device

An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.

AI Inference Edge Device

Problem: An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM and RRAM and recommend an optimized hierarchical memory structure.

Answer

Memory	Speed	Power	Volatility	Main Use
NAND Flash	Medium/slow	Low	Non-volatile	Model storage
DRAM	Fast	Higher	Volatile	Active model/data
RRAM	Fast	Low	Non-volatile	AI model/parameter storage

Proposed Hierarchy

AI Accelerator → SRAM/Cache → DRAM → RRAM → NAND Flash

- Store frequently used AI weights in fast local cache.
- Load active models into DRAM during inference.
- Use RRAM for frequently used models because it is non-volatile and potentially energy efficient.
- Use NAND Flash for large model storage.
- Use model compression and quantization to reduce memory requirements.

Result

The proposed structure provides fast model loading, lower power consumption and efficient AI inference.

5. High-Performance Gaming Console

A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

High-Performance Gaming Console

Problem: A gaming console suffers frame drops during large scene loading. Analyze L3 cache versus DRAM throughput and propose memory hierarchy enhancements.

Answer

L3 cache has much lower latency than DRAM, but its capacity is small. DRAM has much larger capacity and higher bandwidth but greater latency.

Feature	L3 Cache	DRAM
Latency	Very low	Higher
Capacity	Small	Large
Speed	Very high	High
Purpose	Frequently used data	Large game assets

Proposed Improvements

- Increase L3 cache capacity.
- Use multiple memory channels.
- Apply prefetching for upcoming game assets.
- Use DMA for fast data transfers.
- Keep frequently accessed textures and game logic in cache.
- Stream large assets into DRAM before they are required.

Proposed Hierarchy

CPU/GPU → L1 → L2 → L3 → High-Bandwidth DRAM → SSD

Result : The enhanced hierarchy reduces memory stalls and improves scene-loading performance, resulting in fewer frame drops and smoother gameplay.

Code of Conduct:

I (AMARA SATWIK / 192525204) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.SATWIK

MARKS ALLOCATION

	Understanding of Industry Problem (20%)	Application of Engineering Concepts (25%)	Solution Design (25%)	Use of Tools (15%)	Analysis (10%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

